N.S.LABORATORY SDN. BHD.

FOFENICOL, 100mg/ml, Solution for use in drinking water

Active Ingredients:
Each ml contains
Florfenicol---100mg
Product Description:
Clear, colourless to yellow liquid
Dosage Form:
Solution for use in drinking water
Pharmacodynamics:
Florfenicol is a broad-spectrum synthetic antibiotic in the phenicol group that is active against most Gram-positive and Gram-negative bacteria isolated from domestic animals. Florfenicol acts by inhibition of protein synthesis at the ribosomal level and is bacteriostatic. However, bactericidal activity has been demonstrated *in-vitro* against *Actinobacillus pleuropneumoniae* and *Pasteurella multocida* when florfenicol is present at concentrations above the MIC for up to 12 hours. *In-vitro* testing has shown that florfenicol is active against the bacterial pathogens most commonly isolated in respiratory diseases in pigs, including *Actinobacillus pleuropneumoniae* and *Pasteurella multocida*. The MIC90 values of florfenicol against *Actinobacillus pleuropneumoniae* and *Pasteurella multocida* strains isolated in Czech Republic (2015-2016) and United States and Canada (2011- 2015), were determined as 0.5µg/ml, respectively. For *A. pleuropneumoniae* and *P. multocida*, the CLSI breakpoint of resistance for swine respiratory disease is 8µg/ml (2013). Acquired resistance to florfenicol is associated with several genes, including FloR which encodes an efflux pump.
Pharmacokinetics:
After administration to pigs by gavage at 15mg/kg under experimental conditions, absorption of florfenicol was variable but peak serum concentrations of approximately 5µg/ml were reached approximately 2 hours after dosing. The terminal half-life was between 2 and 3 hours. When pigs were given free access, for 5 days, to water medicated with the veterinary medicinal product at a concentration of 100mg florfenicol per litre of water, serum concentrations of florfenicol exceeded 1µg/ml for the entire 5 days treatment period except for a couple of short excursions below 1µg/ml. After absorption and distribution, florfenicol is extensively metabolised by pigs and rapidly eliminated, primarily in urine. After parenteral dosing of florfenicol to pigs, it has been shown that lung concentrations are similar to serum concentrations.
Environmental Properties:
Florfenicol is toxic to cyanobacteria.
Indication:
In pigs:
Treatment and metaphylaxis at the group level where clinical signs are present of swine respiratory disease associated with *Actinobacillus pleuropneumoniae* and *Pasteurella multocida* susceptible to florfenicol. The presence of the disease should be established in the herd before initiating metaphylactic treatment.
Mode of Administration:
Oral administration via drinking water.
Recommended Dosage:
In drinking water use. The recommended dosage is 10mg of florfenicol per kg bodyweight per day in drinking water for 5 consecutive days. The exact daily amount of the veterinary product should be calculated according to the following formula:

$$\frac{\text{X ml veterinary product/ kg b.w./ day} \times \text{Mean body weight (kg) of animals to be treated}}{\text{Mean daily water consumption (litre) per animal}} = \text{X ml veterinary product per litre drinking water}$$

The appropriate quantity of medicated water should be prepared based on the daily water consumption.To ensure a correct dosage body weight should be determined as accurately as possible. In order to avoid under- and over-dosing, treated animals should be divided into groups of similar bodyweight and the dose should be calculated for each group individually.
For Bulk Tank:
To treat pigs drinking 10% of their bodyweight, at the dose of 10 mg/kg: add the florfenicol solution to the drinking water in the bulk tank. Use one bottle (1L) of florfenicol solution for every 1000 L of water or use one barrel (5L) of florfenicol solution for every 5000 L of water and mix thoroughly.
For Proportioner:
To treat 5,000 kg of pigs, drinking 10% of their bodyweight, at the dose rate of 10 mg/kg:
1. Empty the content of one bottle/barrel of florfenicol solution in the proportioner and dilute with drinking water as follows:

Bottle/Barrel	Amount of drinking water
1L	100 L
5L	500 L

2. Mix thoroughly.
3. Set the proportioner on 10%
4. Turn on the proportioner.

Warning: Solutions with concentrations higher than 1.2 g of florfenicol per litre may precipitate.

The uptake of medicated water depends on several factors including the clinical state of the animals and local conditions such as ambient temperature and humidity. In order to obtain the correct dosage water uptake has to be monitored and the concentration of florfenicol has to be adjusted accordingly. If however it is not possible to obtain sufficient uptake of medicated water animals should be treated parenterally.
Contraindications:
Do not use in boars intended for breeding purposes. Studies in rats have revealed evidence of potential adverse effects on the male reproductive system. Do not use in case of hypersensitivity to the active substance or to any of the excipient.
Special Warnings for each Target Species:
The treated pigs should be placed under special observation. On each of the five days of treatment, unmedicated drinking water should not be given until the full daily amount of medicated drinking water has been ingested by pigs. If there are no signs of improvement after three days of the treatment, the diagnosis should be reviewed and, if necessary, the treatment changed.
Special Precautions for Use in Animals:
The veterinary medicinal product should be used in conjunction with susceptibility testing. Use of the product deviating from the instructions given may increase the prevalence of bacteria resistant to the florfenicol. Official and local antimicrobial policies should be taken into account when the product is used. Treatment should not exceed 5 days.
Special Precautions to be taken by the Person Administering the Medicinal Products to Animals:
This product may cause irritation of the skin and eyes. Florfenicol may cause adverse effects on male reproductive systems, such as shrinking of the testes. Contact of the neat product, or the medicated drinking water, with skin and eyes should be avoided. Personal protective equipment consisting of protective gloves, coverall and safety glasses should be worn when handling and mixing the product. Do not smoke, eat or drink when handling the product or mixing the medicated drinking water. In case of accidental spillage into eyes, wash them immediately with water. In case of contact with skin, wash the affected area immediately and remove any contaminated clothing. This product may cause hypersensitivity (allergic) reactions in some people. People with known hypersensitivity to florfenicol or propylene glycol should avoid contact with the veterinary medicinal product. If you develop symptoms following exposure, such as skin rash, seek medical advice and take the package leaflet or the label with you.
Other precautions
This product should not be allowed to enter surface waters as it has harmful effects on aquatic organisms.
Interactions with other Medicaments:
No data available
Usage during Pregnancy and Lactation:
Studies in laboratory animals have not revealed any evidence of potential embryotoxic or foetotoxic effect of florfenicol. The safety of the veterinary medicinal product in sows has not been established during pregnancy and lactation. The use is not recommended during pregnancy and lactation.
Adverse Effects:
A slight reduction of water consumption by the animals, dark brown faeces and constipation may be observed during treatment. Very commonly observed adverse effects are diarrhea and/or peri-anal and rectal erythema/ oedema which may affect approximately 40% of the animals. These effects are transient. In a few of the affected animals, prolapse of the rectum, that resolves without treatment may be observed.
Overdose and Treatment:
After administration at 3 times the recommended dose a decrease in weight gain, food and water consumption, peri-anal erythema and oedema and modification of some haematological and biochemical parameters indicative of dehydration may be observed.
Withdrawal Periods (Meat and Offal):
Pigs- 20 days
Storage Condition:
Store in cool, dark place(< 30ºC).
Shelf Life:
36 months
Shelf Life after Opening:
Discard any unused solution immediately after opening
Shelf Life after Dilution/Reconstitution:
24 hours
Packaging Available:
1L/bottle; 12 bottles/carton; 12kg/carton
Manufacturer and Registration Holder:
N.S. Laboratory Sdn. Bhd.
54, Jalan S2C2, Green Technology Park, Seremban 2, 70300 Seremban, Negeri Sembilan, Malaysia.
Revised Date:
26/03/2024